

DfX Rationale - Team 05

1. Concept Selection Matrix (Pugh Matrix)

Three control strategies were evaluated for the line-follower robot:

- Alternative 1: PID Controller
- Alternative 2: State-Space (Pole Placement)
- Alternative 3: Fuzzy Logic Controller

Selection Criteria and Weights:

Criterion	Weight	Justification
Performance (IAE minimization)	40%	Primary requirement R-TRACK-02 ($IAE < 0.030\text{ m}\cdot\text{s}$)
Tuning Ease	25%	CA2 timeline requires quick implementation
Computational Cost	20%	Real-time feasibility on embedded system
Robustness	15%	Handles disturbances (curves, noise)

Pugh Matrix:

Criterion	Weight	PID	State-Space	Fuzzy Logic
Performance	0.40	8	9	7
Tuning Ease	0.25	9	6	5
Computational Cost	0.20	9	7	6
Robustness	0.15	7	8	8
Weighted Score	1.00	8.35	7.65	6.55

Weight Justification:

Performance was weighted 40% because tracking accuracy is the primary functional requirement (R-TRACK-02: IAE < 0.030 m·s). Tuning ease (25%) is critical for meeting the CA2 deadline. Computational cost (20%) ensures the controller can run in real-time on the digital twin. Robustness (15%) addresses disturbance handling during curves and straight sections.

Conclusion:

PID controller was selected with a score of 8.35, offering the best balance of performance and implementation simplicity. State-space scored 7.65 due to tuning complexity. Fuzzy logic scored lowest (6.55) due to high computational cost and tuning difficulty. PID will be implemented for CA2; state-space may be explored in CA3 if IAE target is not met.

2. Life Cycle Assessment (LCA) Screening

A qualitative LCA screening was performed to identify environmental hotspots across the product lifecycle.

Rank	Component	Lifecycle Stage	Environmental Impact	Severity	Mitigation Strategy
1	Lithium-ion Battery	Raw Material Extraction	Cobalt mining pollution, toxic waste	HIGH	Explore LFP batteries or recycling programs
2	DC Motors	Operation	25% energy loss as heat, electricity consumption	MEDIUM	Select ≥85% efficient motors
3	ABS Plastic Chassis	Manufacturing & Disposal	Petroleum-based, non-biodegradable	LOW-MEDIUM	Consider PLA bio-plastic or recycled PETG

Top 3 Environmental Hotspots:

Component	Lifecycle Phase	Impact Level	Justification
Battery (Lithium-ion)	Raw Material Extraction	HIGH	Cobalt mining has toxic byproducts, ethical concerns, and water pollution

Component	Lifecycle Phase	Impact Level	Justification
Motors (75% efficiency)	Use Phase	MEDIUM	25% energy lost as heat over 75-second operation; contributes to energy consumption
Chassis (ABS Plastic)	End-of-Life	LOW-MEDIUM	Petroleum-based, non-biodegradable, but small material mass (recyclability challenges)

Mitigation Considerations for Future Work:

- Battery: Explore Lithium Iron Phosphate (LFP) alternative with lower environmental impact
- Motors: Select high-efficiency model ($\geq 85\%$) to reduce energy loss during operation
- Chassis: Consider PLA bio-plastic or recycled PETG for sustainability

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